

TB-500

Thymosin Beta-4 Fragment | 5mg & 10mg x 10 Vials

Tissue Repair | Angiogenesis | Anti-Inflammatory | SubQ / IM

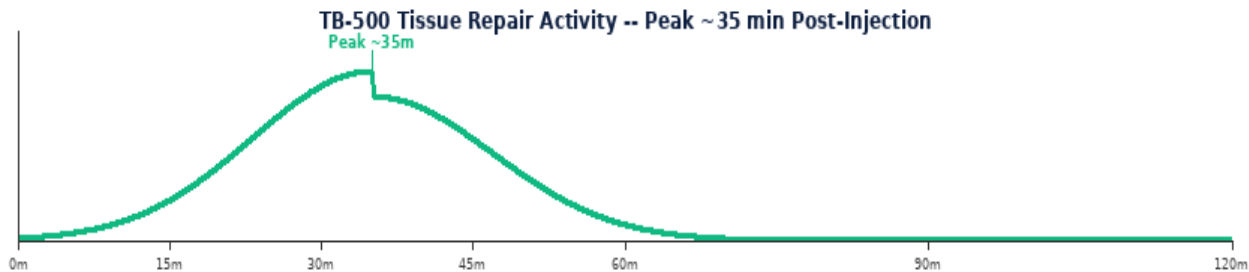
Purity: 99.8% | Endotoxin-Free | Research Use Only

TB-500 is a synthetic fragment (amino acids 17-23) of Thymosin Beta-4 (TB4), the most abundant intracellular peptide in mammalian cells. It promotes tissue repair, angiogenesis, anti-inflammation, and cell migration. TB-500 retains the primary biological activity of full TB4 while offering superior solubility and stability. Key applications: injury recovery, wound healing, cardiac and neurological repair research.



Parameter	Specification
Class	Thymosin Beta-4 fragment (AA 17-23) -- actin-sequestering, tissue repair
Vial Sizes	5mg x 10 10mg x 10 Purity 99.8%
Reconstitution	1 mL BAC Water: 5mg->5mg/mL 10mg->10mg/mL
Standard Dose	500-2,500 mcg per injection SubQ or IM
Frequency	2-3x per week SubQ/IM no fasting required
Key Benefits	Tissue repair, angiogenesis, anti-inflammatory, cardiac/neural repair, actin regulation
Cycle & Storage	4-12 weeks on, 2-4 weeks off 2-8 deg C use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Actin Regulation	Sequesters G-actin monomers; promotes actin polymerisation in migrating cells; enables cell migration to injury sites; central to wound healing and tissue repair initiation	Wound healing; tissue repair; injury recovery
Angiogenesis & Repair	Upregulates VEGF and other angiogenic factors; promotes new blood vessel formation at injury sites; stimulates keratinocyte and fibroblast migration; accelerates re-epithelialisation	Vascular repair; chronic wound models; muscle/tendon healing
Anti-Inflammatory & Cardiac	Downregulates NF-kB inflammatory cascade; reduces TNF-alpha, IL-1beta; promotes cardiomyocyte survival after ischaemia; neural repair via oligodendrocyte progenitor activation	Anti-inflammatory; cardiac protection; neurological repair

Benefit	Context
Injury Recovery	TB-500 accelerates healing in muscle, tendon, ligament, and skin by promoting cell migration, angiogenesis, and anti-inflammatory resolution.
Cardiac & Neural	Research demonstrates TB4/TB-500 promotes cardiomyocyte survival post-ischaemia and activates neural repair pathways.
Synergy with BPC-157	TB-500 and BPC-157 are frequently co-administered for synergistic tissue repair -- complementary mechanisms of action.

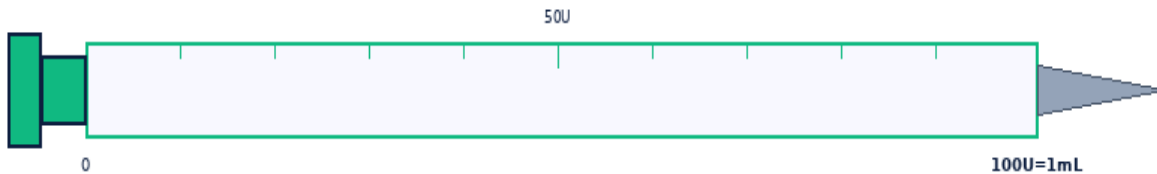
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

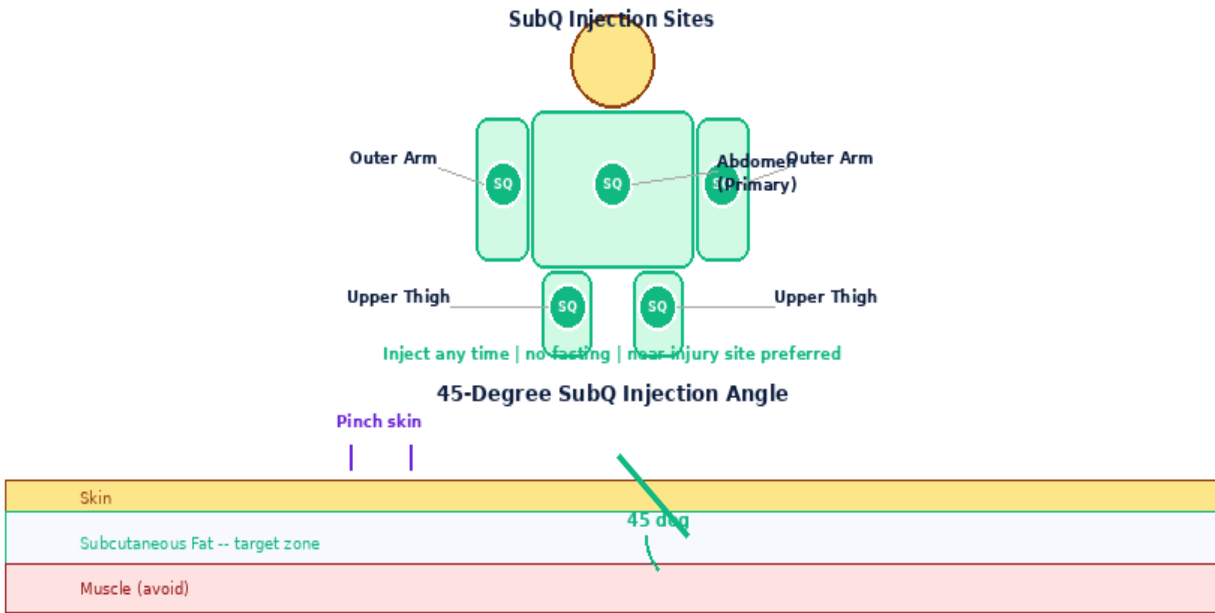
5mg x 10 -- 1mL BAC -> 5.0mg/mL (5,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	2500mcg
Vol(mL)	0.1000	0.2000	0.4000	0.5000
Units	10.0U	20.0U	40.0U	50.0U
Doses/Vial	10	5	2	2

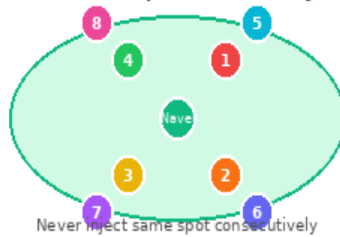
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	2500mcg
Vol(mL)	0.0500	0.1000	0.2000	0.2500
Units	5.0U	10.0U	20.0U	25.0U
Doses/Vial	20	10	5	4

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G syringe. No fasting required.
2	Draw dose	Draw volume per dose table. Inject near injury site if possible.
3	Inject SubQ/IM	SubQ abdomen (systemic) or near target tissue. IM for deeper delivery.
4	Rotate & dispose	8-point rotation. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Acute Injury	2,000-2,500 mcg	2-3x/week first 2 wk	Then reduce	Higher loading dose for acute injury phase.
Maintenance / Prevention	500-1,000 mcg	2x/week	4-12 wk	Lower maintenance dose.
+BPC-157 Stack	TB: 2mg BPC: 500mcg	2-3x/week	4-8 wk	Synergistic tissue repair stack.

Side Effect	Likelihood	Management
Injection site reaction	Common	Mild redness/itch. Rotate sites.
Fatigue (initial)	Occasional	Transient. Resolves in first week.
Headache	Rare	Dose-related. Reduce dose.

Contraindications: Cancer history, pregnancy, under 18.

1. 5mg: 1mL BAC->5mg/mL 10mg: 1mL->10mg/mL.
2. No fasting required. Inject any time.
3. Inject near the target injury site for localised repair benefit.
4. Loading phase: 2,000-2,500 mcg 2-3x/week for first 2-3 weeks.
5. Maintenance: reduce to 500-1,000 mcg 1-2x/week after loading.
6. Stack with BPC-157 for synergistic tissue repair research.
7. Rotate injection sites through 8-point abdomen map.
8. Store 2-8 deg C. Use within 28 days. Never freeze.
9. Contraindicated in active cancer history. Not for pregnancy or under 18.

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